

PROJECT AND TEAM INFORMATION

Project Title

InternMatch: A Real-Time Internship Recommendation System using DSA and OOP

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

- While searching for internship opportunities, students have to go through messy databases.
- Manually checking eligibility criteria of each internship is time-consuming and hard leaving unsure which internship fits their skill.
- InternMatch solve this by checking eligibility and calculating match percentage on basis of skill while also showing missing skill.

State of the Art / Current solution

- Students visit to platforms like LinkedIn, which don't show individual student eligibility or provide a skill match percentage.
- Our project uses of DSA and OOP concepts , to fetch real-time data from public API to check eligibility, calculate match percentage, and identify missing skills.

Project Goals and Milestones

Project Goal: The aim of InternMatch is to create a system that fetches real-time internship data using API, takes student details like- (CGPA, branch, skills), checking eligibility, calculating match percentage, and shows best matching internships.

Phase 1 (Ongoing):

- Finalise requirements and workflow; design Student and Internship class.
- Real-time internship data fetching using public API and handle it using JSON.

Phase 2:

- Store internship data in Linked List; and checked eligibility based on CGPA and branch.
- Implement skill searching/matching, calculating match percentage and skill gap, and do sorting.

Phase 3:

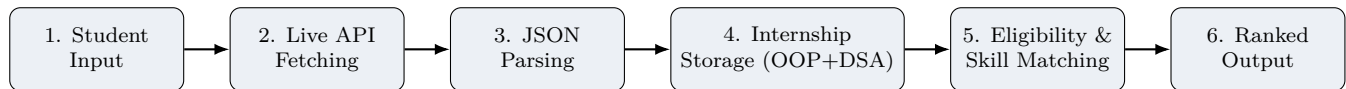
- Internship is recommended on the basis of match percentage and show the missing skill as output.
- Many student profile are test then project posted on github.

Project Approach

- Project made in C++ by OOPS and DSA'S concepts by creating separate student and internship class storing and manage data.
- We fetch real internship data from API using libcurl and handle with nlohmann/json, store in link list.
- The system check eligibility on basis of Cgpa and branch then calculate the match percentage by compare skills and sort internship by match percentage.
- We will use Vs Code and Github to store code and document.

System Architecture (High Level Diagram)

of your choice).



The system take student input (CGPA, branch, skills), fetched live internship data using Arbeitnow API using libcurl, process it with nlohmann/json, store record using a link list (OOP+DSA), check eligibility and match skills, and presented ranked list of best-matched internships.

Project Outcome / Deliverables

- Our project is based on C++ that inputs Student Data and will fetch internship data using public API(arbeitnow.com).
- Eligibility is checked on the basis of CGPA and Branch and skills are compared to calculate match percentage and show missing skills and displays top internships.
- The project working will be maintained on Github.

Assumptions

- We assume a stable internet connection from user side because we are using API to fetch real time data.
- Since the API doesn't consist informations like CGPA and Branch,therefore we use self defined tags to check match quality.

References

1. libcurl Documentation – <https://curl.se/libcurl/>
2. nlohmann/json Library – <https://github.com/nlohmann/json>
3. Arbeitnow Job Board API – <https://www.arbeitnow.com/api/job-board-api>